Roll No. | | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-2)

BIOCHEMISTRY

Subject Code : BP-203T

M.Code : 74969

Date of Examination : 06-05-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION - A

1. **Answer briefly :**

   a) What is amphoteric nature of amino acid?

   b) What is the difference between nucleotide and nucleoside?

   c) What is the difference between $\alpha 1-4$ and $\beta 1-4$ linkage in polysaccharide?

   d) Differentiate between AMP and cAMP.

   e) Differentiate between NADPH and NADH.

   f) What is role of Phosphoglucomutase?

   g) What is the biochemical significance of the pentose phosphate pathway?

   h) What is phenylketonuria?

   i) What is ketogenesis?

   j) Name two essential fatty acids.

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SECTION - B

2. a) Give IUB classification of enzyme.

   b) Explain Lineweaver - Burk plots of enzyme kinetics.

3. Explain the TCA cycle with reactions and its significance.

4. a) Explain semi conservative model of DNA replication.

   b) Give salient features of Watson and Crick model of DNA.

SECTION-C

5. What are primary and secondary structures of protein?

6. Classify energy-rich molecules with examples.

7. Describe the Electron Transport Chain (ETC) with a neat diagram.

8. Differentiate between oxidative phosphorylation and substrate level phosphorylation.

9. Describe biosynthesis of vit D from cholesterol.

10. Describe synthesis and biological significance of Melatonin.

11. Discuss urea cycle and its importance.

12. Write short notes on inhibitors of protein synthesis.

13. Discuss the diagnostic applications enzymes.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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